

Samuel Chau

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Summary

Computer Science graduate and lead maintainer of Profilarr, an open-source configuration-as-code tool for self-hosted media servers (3.5M+ downloads, 2,400 stars). Full-stack web developer with a strong frontend sensibility, seeking frontend or full-stack roles building products people actually like using.

Education

University of Adelaide

Adelaide, SA

Bachelor in Computer Science, Major in Cybersecurity

Feb 2022 – Dec 2025

- GPA: **6.4/7.0** — Achieved 13 High Distinctions
- Top Results: Databases & Ethical Data (96), Web & Database Computing (95), Algorithm & Data Structure Analysis (93), Software Engineering & Project (92)

Experience

Maintainer & Lead Developer

Apr 2023 – present

Profilarr

Open-source tool that brings configuration-as-code to self-hosted media servers.

- Designed a **CRDT-inspired, event-sourced configuration system** — config is stored as operations rather than final state, so upstream curators and local users can edit independently and merge without conflict; value guards catch same-field edits before they silently overwrite.
- Architected and solo-built a **100K+ line SvelteKit/Deno/TypeScript/SQLite application** with a fully type-safe data layer generated from the database schema and queried through Kysely.
- Built a **Terraform-style sync engine** — idempotent, retry-safe, with drift detection — backed by a crash-recoverable job queue.
- Built a reusable **component library** that leverages semantic CSS tokens to create a consistently styled, themable, and polished design, enabling rapid UI development. Enforced using custom Svelte-AST lint rules that ban raw HTML and hardcoded styles.
- Designed a **defense-in-depth security model that keeps raw secrets off the web surface entirely**: clients receive presence flags instead of values, API keys are hashed like passwords, and backups are credential-stripped before download. Hardened against OWASP Top 10 (sanitized markdown, CSRF, path-traversal, rate limiting).
- Built a **CI/CD release pipeline** running build, lint, type checks, unit + integration tests against the production binary, Semgrep SAST, and Playwright E2E auth flows on every change.
- Grew to **3.5M+ downloads and 2,400 GitHub stars** ran all support, beta coordination, and contribution guidelines for a 2,000+ member community.

Projects

Synopsis

React, Express, LLMs, FFmpeg

AI Research Paper Video Platform

- Built full-stack platform using Grobid for paper parsing, LLMs for summarization/image generation, and FFmpeg for video synthesis.
- Pitched at **Tech eChallenge** startup competition; ranked **Top 5** out of 50+ teams.

OMesh

Python, Docker, WebSockets

Secure Messaging Platform

- Built end-to-end encrypted messaging platform using low-level socket programming and custom protocol design.
- Implemented RSA/AES-GCM cryptographic pipeline for secure key exchange and message encryption.
- Deployed multi-server architecture with Docker Compose; real-time communication via WebSockets.

Rogue

C++, SFML

Procedural Roguelike Game

- Built turn-based roguelike with procedural map generation and entity component system architecture.
- Implemented design patterns (Factory, Observer, State) for extensible game object behavior.

Skills

Languages: TypeScript, Python, C++, C#, SQL, HTML/CSS

Technologies: SvelteKit, Deno, Docker, Git, Linux, SQLite, Kysely, Discord.js, FFmpeg

Practices: GitOps, Event Sourcing, API Design, Open Source Development

Interests: Self-Hosting, Homelabbing, Media Archival, Basketball, Technical Writing